TONY WILBERG

AI PLATFORM ENGINEER · APPLIED AI / MLOPS · 20 YEARS IT OPERATIONS

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PROFESSIONAL SUMMARY

Self-directed engineer who builds production-grade AI systems and operates them end to end, backed by twenty years running real IT operations across healthcare, K–12 education, and medical-device manufacturing. Designed and runs a multi-host, multi-agent AI platform: intent-based agent orchestration with shadow-to-live cutover and instant rollback, a Retrieval-Augmented Generation service over a vector database, and a Model Context Protocol (MCP) layer exposing tools to agents and IDE clients. Pairs hands-on infrastructure depth (Linux services, Docker, identity and access, fleet lifecycle, OS migrations, ERP go-lives) with applied-AI engineering (LLM APIs, multi-agent orchestration, retrieval, model-selection evaluation, cost-aware routing). Three relevant AAS degrees — Cybersecurity, Information Systems, Networking (Summa Cum Laude) — plus Azure certification. Seeking AI platform / MLOps / AI solutions-implementation roles where building the system and running it in production are the same job.

TECHNICAL SKILLS

AI / ML Engineering: LLM application development (Anthropic Claude, OpenAI, local models via Ollama); multi-agent orchestration & intent-based routing; Retrieval-Augmented Generation (RAG); vector databases (ChromaDB); Model Context Protocol (MCP) servers & clients; prompt engineering & system-prompt versioning; model-selection evaluation & benchmarking; cost-aware model routing; least-privilege agent tool design

Languages & Automation: Python, PowerShell, Bash; Git

Infrastructure & Platform: Linux (Ubuntu), systemd service management, Docker, VMware; Tailscale; multi-host service deployment; Open WebUI

Cloud & Identity: Microsoft Azure (AZ-900; AZ-104 in progress), Microsoft Entra ID (Azure AD), Active Directory, Group Policy, Microsoft 365

IT Operations: Endpoint & fleet lifecycle, MDT imaging, Jamf Pro MDM, BitLocker key management, OS migration (Win 7→10→11), ITIL / SLA, Jira Service Management

Operating Systems: Windows, macOS, Linux, iOS, Android

SELECTED AI / ML ENGINEERING PROJECTS (INDEPENDENT)

DreamBox — Multi-Agent AI Platform *· 2025–present*

A self-built, production-style platform that hosts AI agents, routes work between them, and runs as a coordinated multi-host stack — the centerpiece of an end-to-end applied-AI system.

- Designed and operate a multi-agent orchestration layer: intent-based routing dispatches specialized agents (knowledge-base curation, code review) on commit events, with a shadow-to-live cutover mechanism and instant one-flag rollback.
- Enforced least-privilege agent design — per-agent tool allowlists plus tier / egress / output guards — so an agent can only call the tools and reach the surfaces its manifest declares, with human-in-the-loop review before any output is published.
- Built a RAG service over a ChromaDB vector store (~117 curated knowledge articles) and an MCP bridge exposing semantic search and verbatim document retrieval to agents and IDE clients.
- Ran model-selection evaluations (local 9B / 27B models vs. a frontier model) for agent tasks and built a cost-aware pattern: a substantive-change filter prevents model loads on routine bookkeeping commits to avoid GPU contention, with local generation gated by a frontier-model reviewer.
- Operate the platform as six Linux / systemd services plus containerized components (vector DB, web UI, TTS, search) across multiple Tailscale-networked hosts, with a synthetic-credential sandbox that structurally prevents secret leakage during development.

TradeBot — Autonomous Trading System *· paper-trading*

- Built an autonomous, multi-asset (futures / options / equities) trading system with a disciplined paper-first validation posture — strategy evaluation, backtesting, and thesis development before any live deployment or capital at risk.
- Engineered the execution path to remove emotion from trade timing and to validate strategies in a live paper environment ahead of a deliberate, gated go-live decision.

Dereth — Machine-Learning Game Agent *· ML / computer vision / navigation*

- Built an autonomous agent that reads live game state and drives combat, looting, and navigation through ML-based decision-making — authoring the decision engine itself rather than scripting fixed plug-in paths.

- Developing a SLAM-based spatial-mapping and computer-vision navigation pipeline with an ML decision layer for task and quest selection.

PROFESSIONAL EXPERIENCE

IT Support Specialist — MicroAire Surgical Instruments, Charlottesville, VA *2021–2025*

- Led the companywide Windows 10→11 migration across all production workstations in a manufacturing environment where downtime carries direct production cost — owning software validation and endpoint verification.
- Primary onsite IT resource through a full ERP transition (Wintergrate-Dataflo → Epicor): coordinated hardware readiness and resolved workstation and software conflicts throughout go-live.
- Re-architected the MDT imaging environment from a single universal image into device-type-specific images after driver conflicts emerged — improving deployment reliability and cutting post-image remediation time.
- Ran a systematic RAM audit across 300+ endpoints ahead of ERP go-live, using a matched-pair methodology to hit performance targets without blanket procurement. Earned AZ-900 (2022).

IT Support Specialist — Entegee (contract to MicroAire), Charlottesville, VA *2020–2021*

- Sole onsite technician throughout the COVID-19 period; led the companywide Windows 7→10 migration with software validation across all endpoints.

Service Desk Team Lead — Richmond Public Schools, Richmond, VA *2015–2019*

- Promoted to Team Lead within one year; owned all high-priority escalations and district-wide IT communications.
- Diagnosed recurring account-creation failures during the Rapid Identity rollout by tracing them to upstream automation logic, then coordinated a fix that prevented recurrence across all affected schools.
- Drove the selection, configuration, and ground-up deployment of Jamf Pro MDM for Apple devices district-wide.

Service Desk Analyst II — Mary Washington Healthcare, Fredericksburg, VA *2011–2013*

- Independently audited 600+ Altiris tickets generating SQL errors, identified the data-loss pattern, corrected it, and recovered 600+ lost service records — eliminating the recurring failure and restoring full SLA visibility. Earned HDI Certification (98%, 2012).

Computer Technician (CTT) — Stafford County Public Schools — Colonial Forge HS, Stafford, VA *2005–2011*

- Administered the full identity and access lifecycle for 2,200+ students and 220+ staff — account creation, application rights, folder permissions, provisioning, and expiration — running the environment solo for most of a six-year tenure, with role-based access across 100+ applications and 1,000+ endpoints at zero unplanned downtime.

Earlier: Service Desk Analyst, Robert Half Technology / Richmond Public Schools (2014–2015) · Technical Support Specialist, Prince William County Public Schools (2011).

EDUCATION

Germanna Community College — *Summa Cum Laude*

AAS, Cybersecurity · AAS, Information Systems Technology (2023) · AAS, Networking (2017)

Career Studies Certificates: Cybersecurity, Data Center Technician, Networking, Advanced Networking

CERTIFICATIONS

AZ-104 Azure Administrator (in progress) · AZ-900 Azure Fundamentals (2022) · HDI Support Center Analyst (2012)